

RC电路

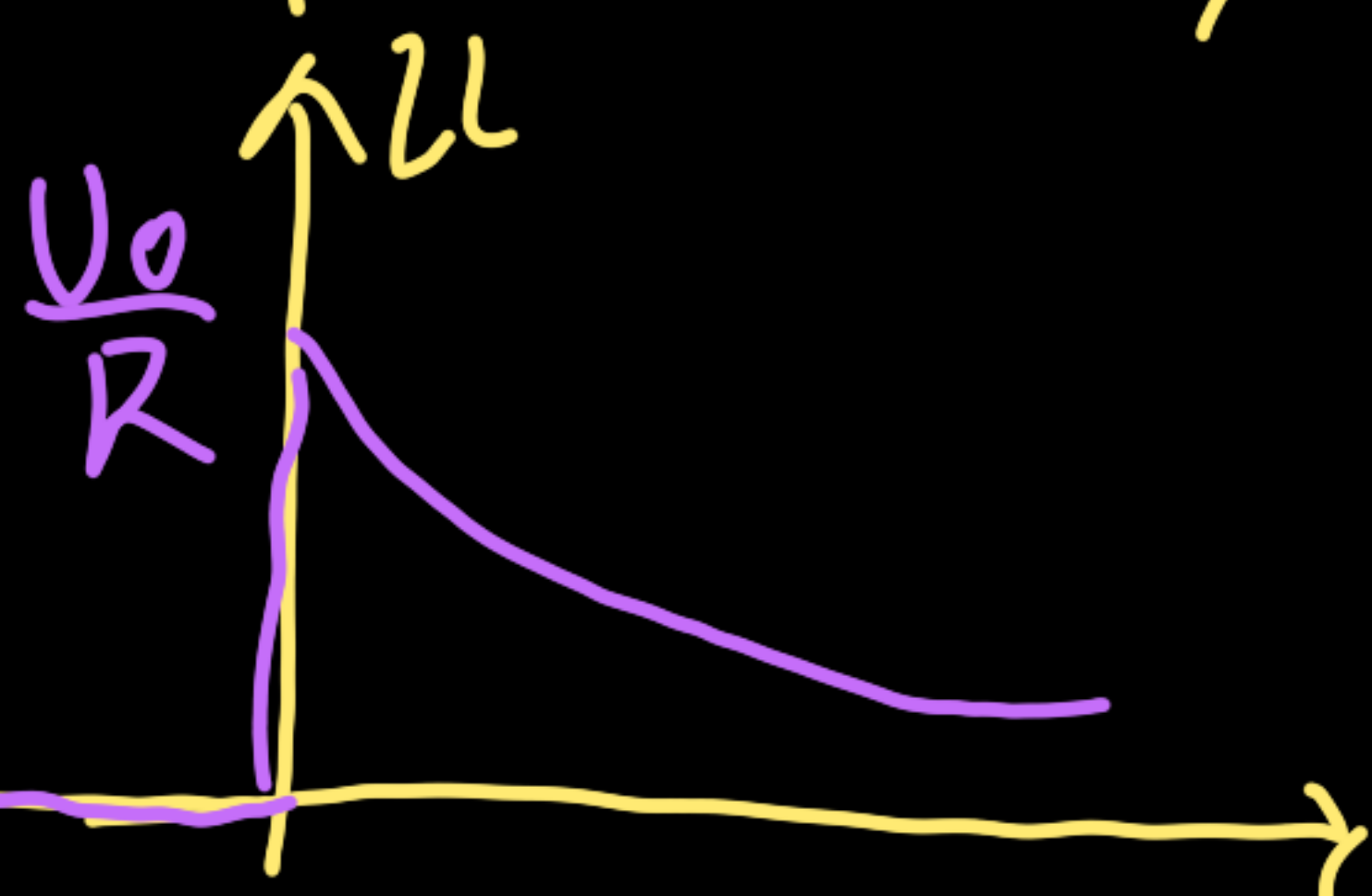
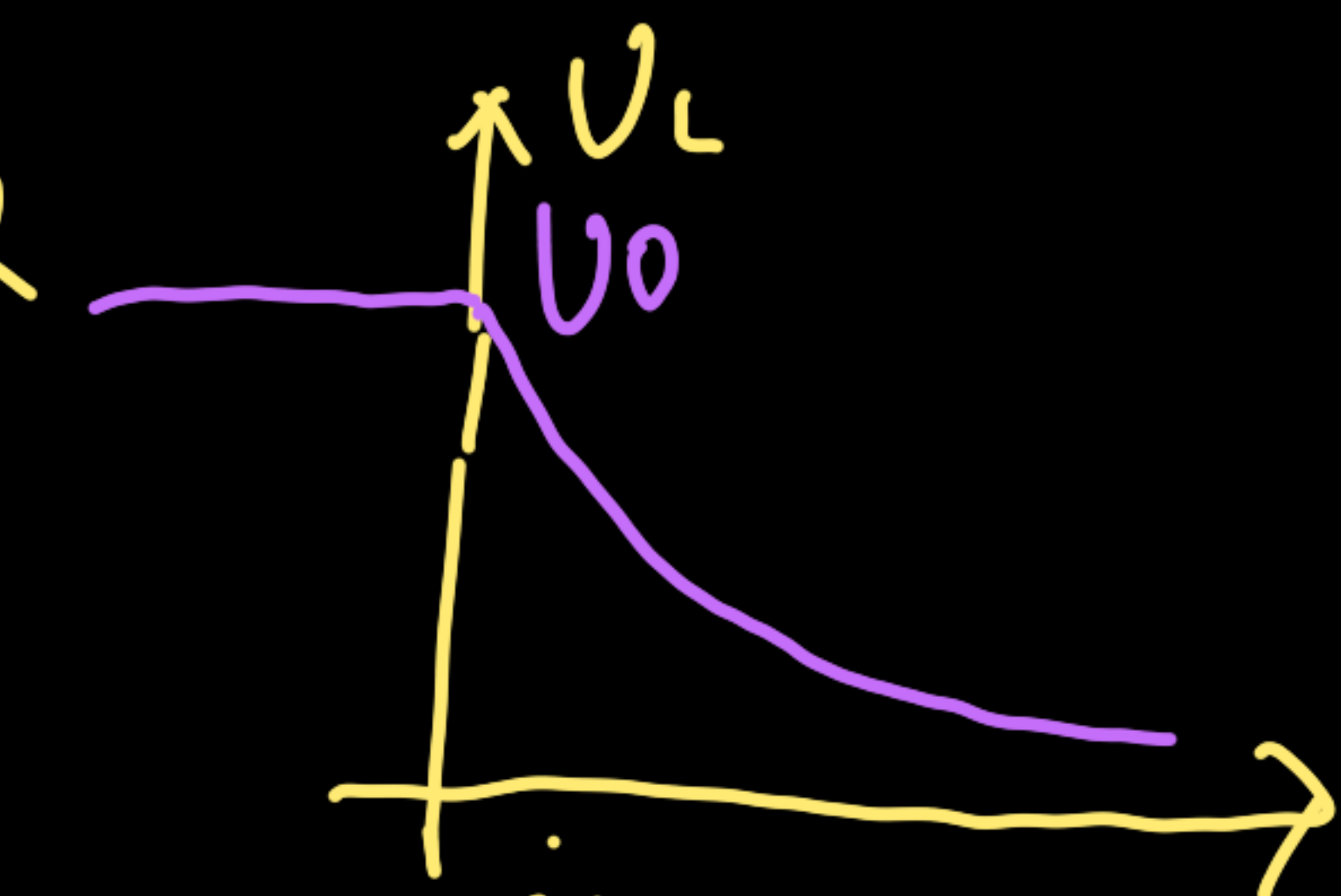
零输入

$$U_C(0_-) = 0$$

$$U_C = U_0 e^{-\frac{t}{RC}}$$

$$I_C = \frac{U_0}{R} e^{-\frac{t}{RC}}$$

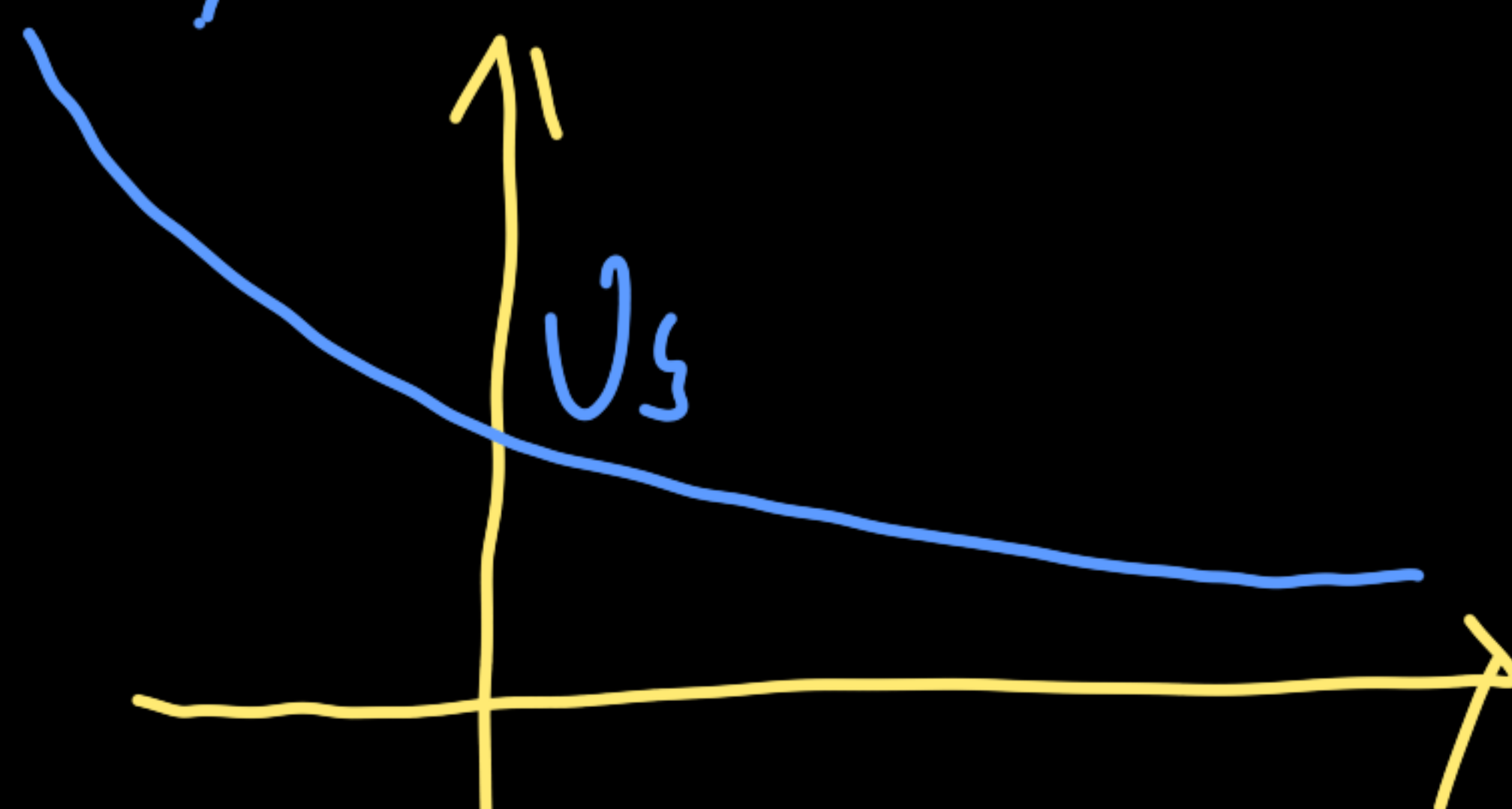
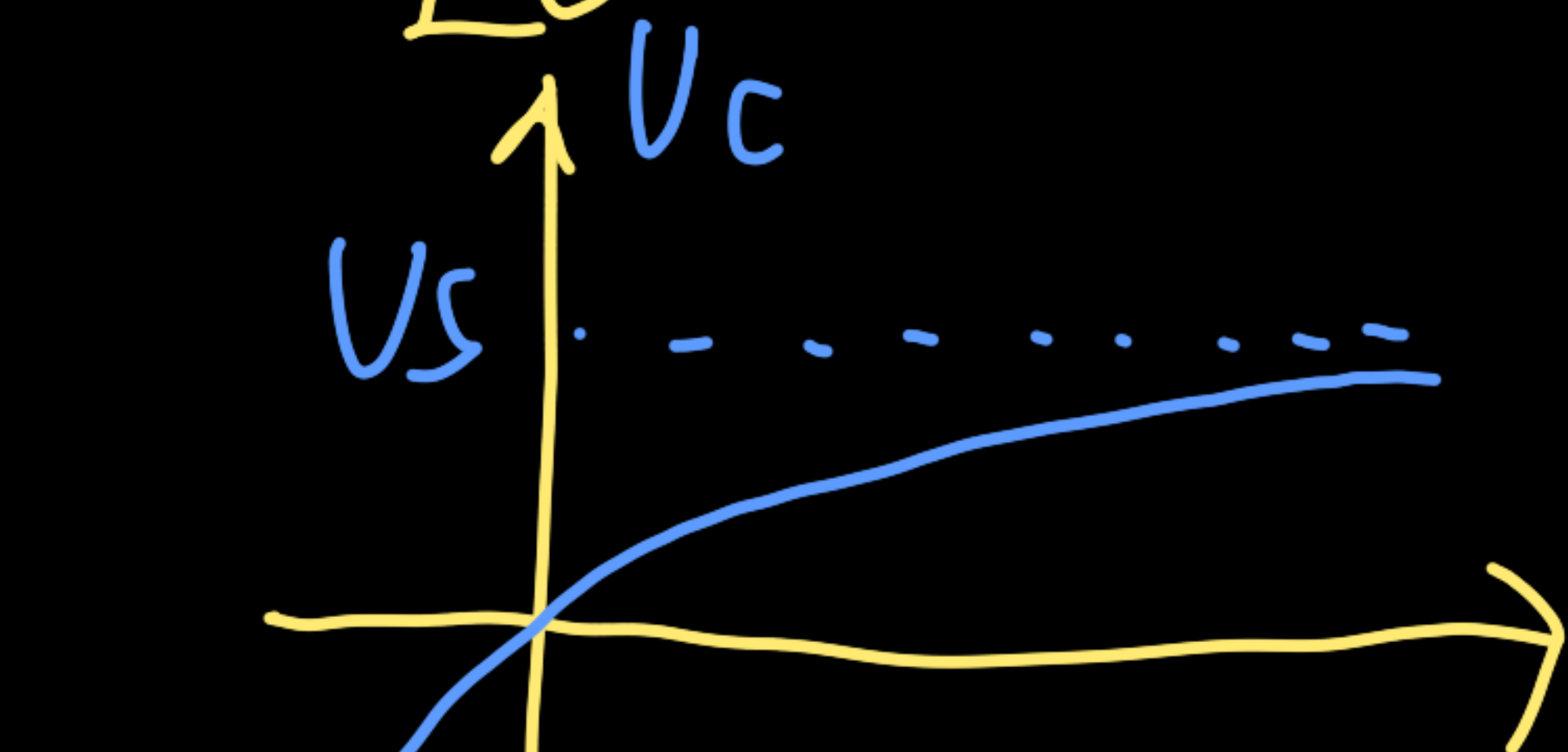
U_C 图像



零状态

$$U_C = U_S (1 - e^{-\frac{t}{RC}})$$

$$I_C = U_S e^{-\frac{t}{RC}}$$



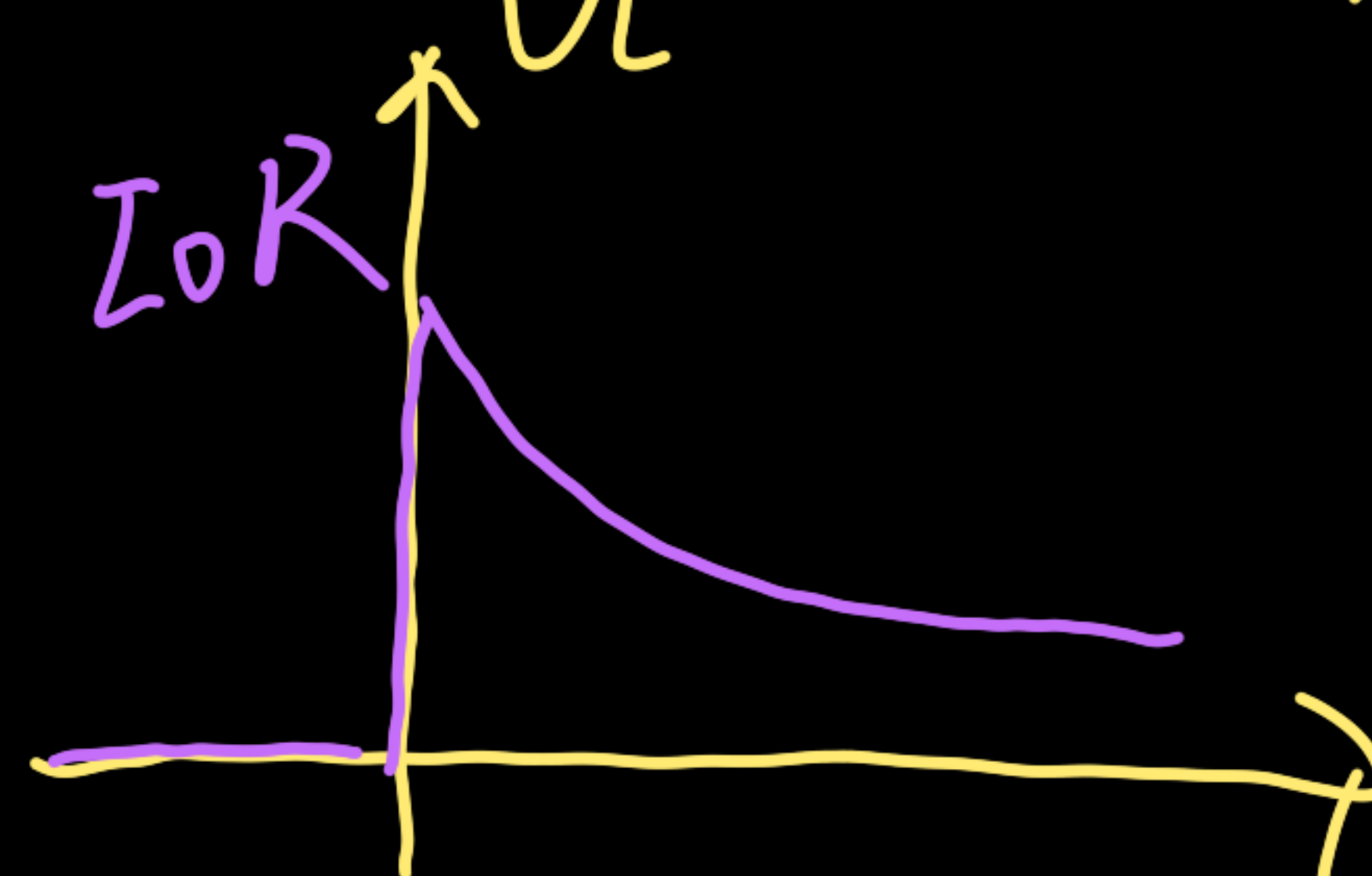
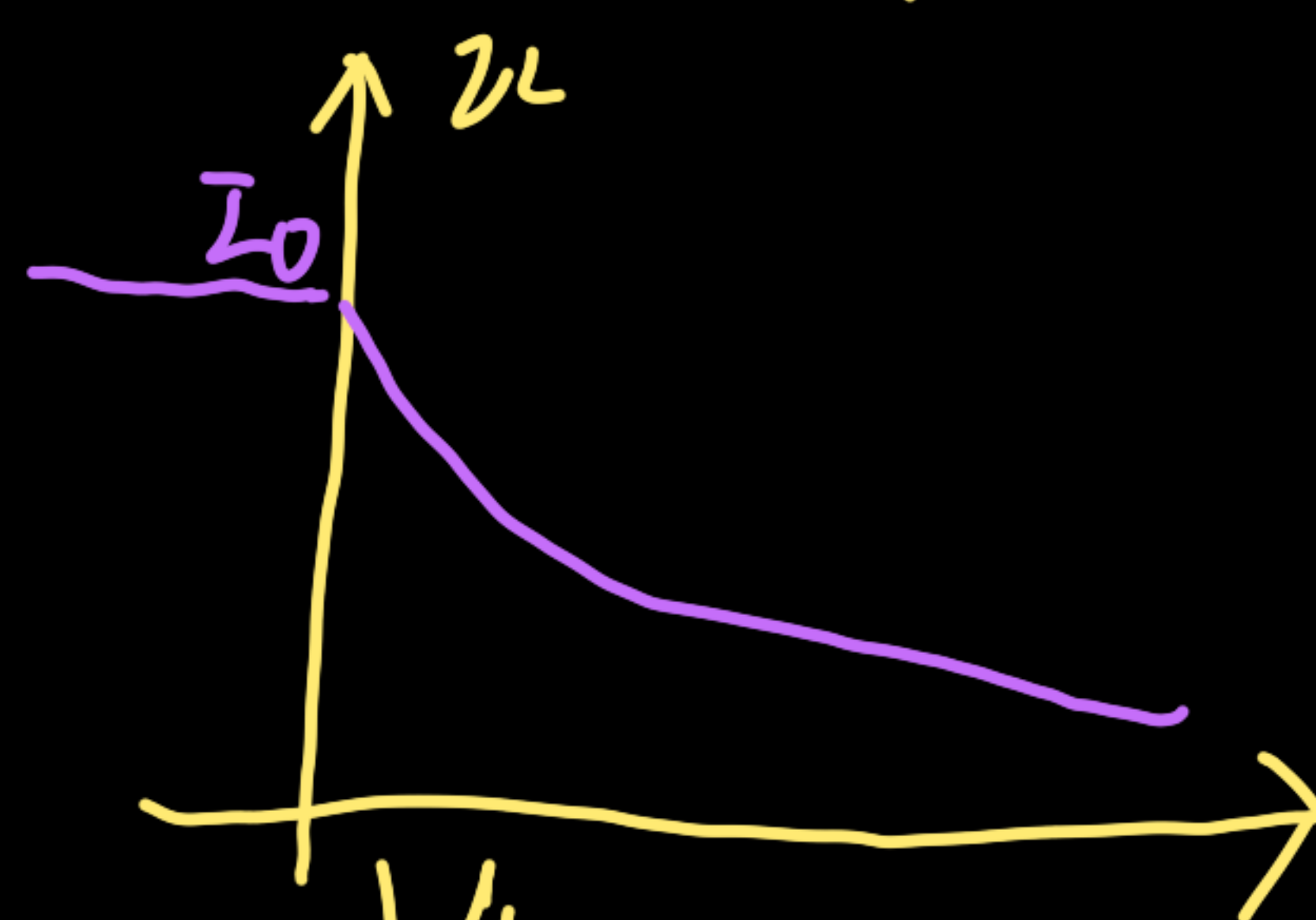
$$U_C = U_S (1 - e^{-\frac{t}{RC}}) = U(\infty) (1 - e^{-\frac{t}{RC}})$$

RL电路

$$i_L(0_-) = 0$$

$$i_L = I_0 e^{-\frac{L}{R}t}$$

$$U_L = I_0 R e^{-\frac{L}{R}t}$$



$$i_L(t) = \frac{U_S}{R} (1 - e^{-\frac{L}{R}t})$$

$$U_L(t) = U_S e^{-\frac{L}{R}t}$$

